

Minimum Propulsion power

following MEPC/Res.232(65) '2013 Interim guidelines for determining minimum propulsion power to maintain the manoeuvrability of ships in adverse conditions'

Ship Data

Title	Data	Comment
DNVGL	n/a	
IMO Number	9123123	
Vessel Name	Titanic Sample	
Ship Type	Bulk Carrier	
Building Yard	Mighty Dry Dock	
Hull Number	ABXXX	

Environmental Factors

Parameters	Value	Unit	Comment
Density of Air (air)	1.225	kg/m ³	
Density of Water (H2O)	1025.8	kg/m ³	
Temperature (Temp)	15	°C	
Viscosity of Water (H2O)	1.184	m ² /s	
Gravity (g)	15	m/s ²	

Principal Particulars

Parameters	Value	Unit	Comment
Length Between Perpendiculars (LPP)	225	m	
Breadth on water line (BWL)	32	m	
Scantling draft at midship (Tm)	14.45	m	
Light Ship Weight	65432	ton	
Deadweight Tonnage (DWT)	82025	ton	
Block Coefficient (BlockC)	0.884		
Wetted surface area (S)	8591	m ²	
Frontal Windage Area (FWA)	905	m ²	
Lateral Windage Area (LWA)	578	m ²	
Rudder Total Area (AR)	51	m ²	

Main Engine Particulars

Parameters	Value	Unit	Comment
Number of Engines	1		
Manufacturer	PowPower Co. Ltd		
Engine Type	Titanic Super 9000MXZ		
Maximum Continuous Rating (MCR)	9801	kW	
RPM of main propulsion engine at MCR after installation of EPL	81.7	rev/min	

Propeller Particulars

Parameters	Value	Unit	Comment
Propeller Type	FPP		
Number of blades	5		
Diameter (Dp)	7.2	m	

Assessment Level 1 – Minimum Power Line

Parameters	Value	Unit	Comment
DWT	82025	ton	
MCR	9801	kW	
a	0.0763		
b	3374.3		

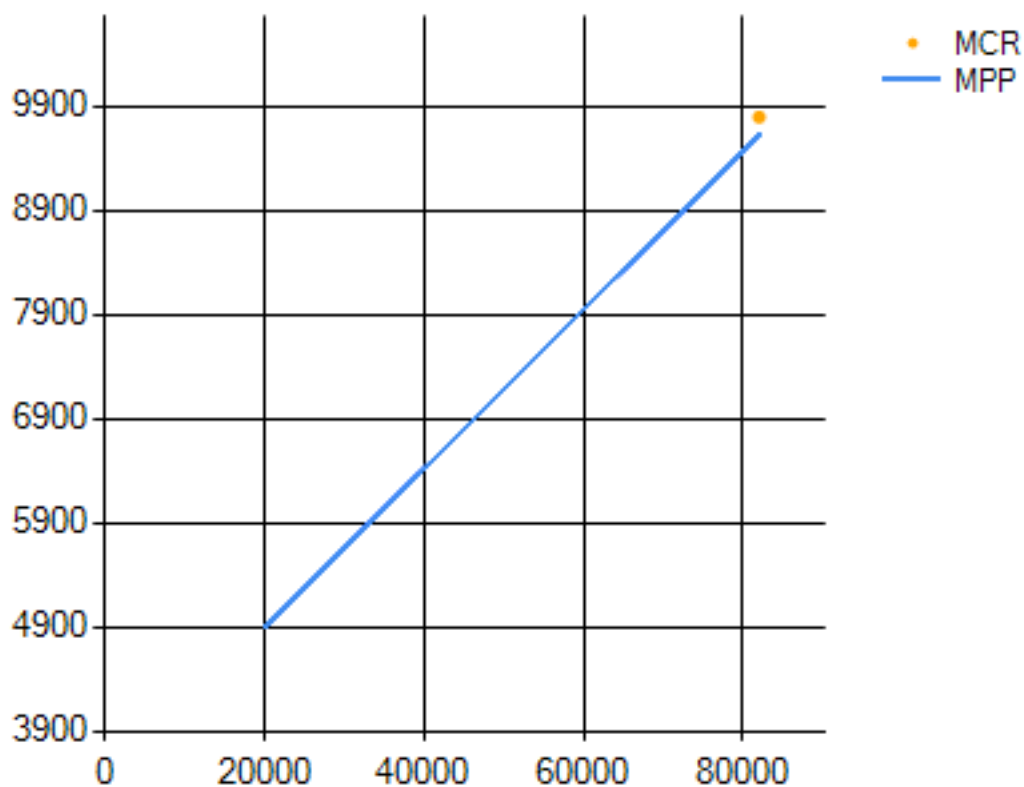
Minimum Power Line Value

$$= a \times (\text{DWT}) + b$$

$$= 0.0763 \times 82025 + 3374.3$$

$$= 9633 \text{ kw}$$

The relationship between MCR and MPP can be shown in the graph below



The requirement according to assessment level 1 is fulfilled.

Assessment Level 2 – Simplified Assessment

Parameters	Value	Unit	Comment
Breadth on water line (BWL)	32	m	
Scantling draft at midship (Tm)	14.45	m	
Length Between Perpendiculars (LPP)	225	m	
Rudder Total Area (AR)	51	m ²	

Using the above parameters, ALScor and % AR / ALS,cor can be calculated.

Submerged lateral area of ship corrected for breadth effect, (ALScor)	4895	m ²	
Percentage of Rudder area to submerged lateral area of the ship (%AR/ALS,cor)	1.042	%	

Now the required speed of advance (Vs) can be computed.

Parameters	Value	Unit	Comment
Frontal Windage Area (FWA)	905	m ²	
Lateral Windage Area (LWA)	578	m ²	
Ratio	1.566		
Minimum Navigation Speed (Vnav)	4	m/s	
Reference Course Keeping Speed (Vck,ref)	4	m/s	
Minimum Course Keeping Speed (Vck)	2.582	m/s	
Required Advance Speed,(Vs)	2.058	m/s	

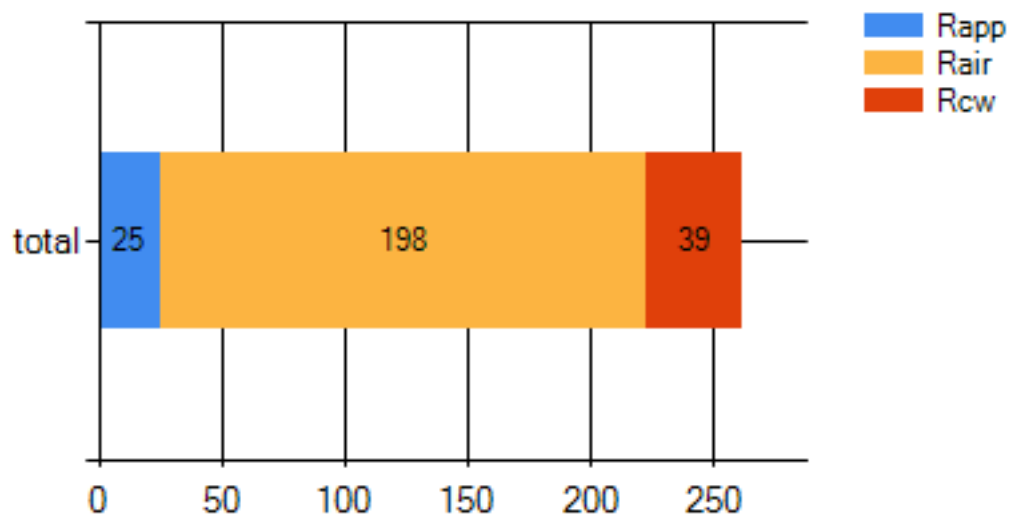
The required speed of advance (Vs) is 2.058 m/s

In the procedure of assessment of installed power, the total resistance can be found.

Parameters	Value	Unit	Comment
No. of Engines	1		
Block Coefficient (BlockC)	0.884		
Temperature (Temp)	15	°C	
Density of Air (air)	1.225	kg/m ³	
Viscosity of Water (H2O)	1.184 x 10 ⁽⁻⁶⁾	m ² /s	
Density of Water (H2O)	1025.8	Kg/m ³	
Wetted area of bare hull (S)	8591	m ²	

Form factor (k)	0.2126		
Reynold Number (Rn)	3910.47292×10^5		
Mean wind speed (Vw)	17.35	m/s	
Frictional Resistance Coefficient (Cf)	0.001726		
Resistance in calm water (Rcw)	39.047	kN	
Aerodynamic resistance (Rair)	198.349	kN	
Aerodynamic resistance coefficient (Cair)	0.95		
Resistance due to appendages (Rapp)	25	kN	

Below is the stacked bar chart to illustrate the subtotal for Rcw + Rair + Rapp.



Subtotal for Rcw + Rair + Rapp
 $= 39.047 + 198.349 + 25$
 $= 262 \text{ kN}$

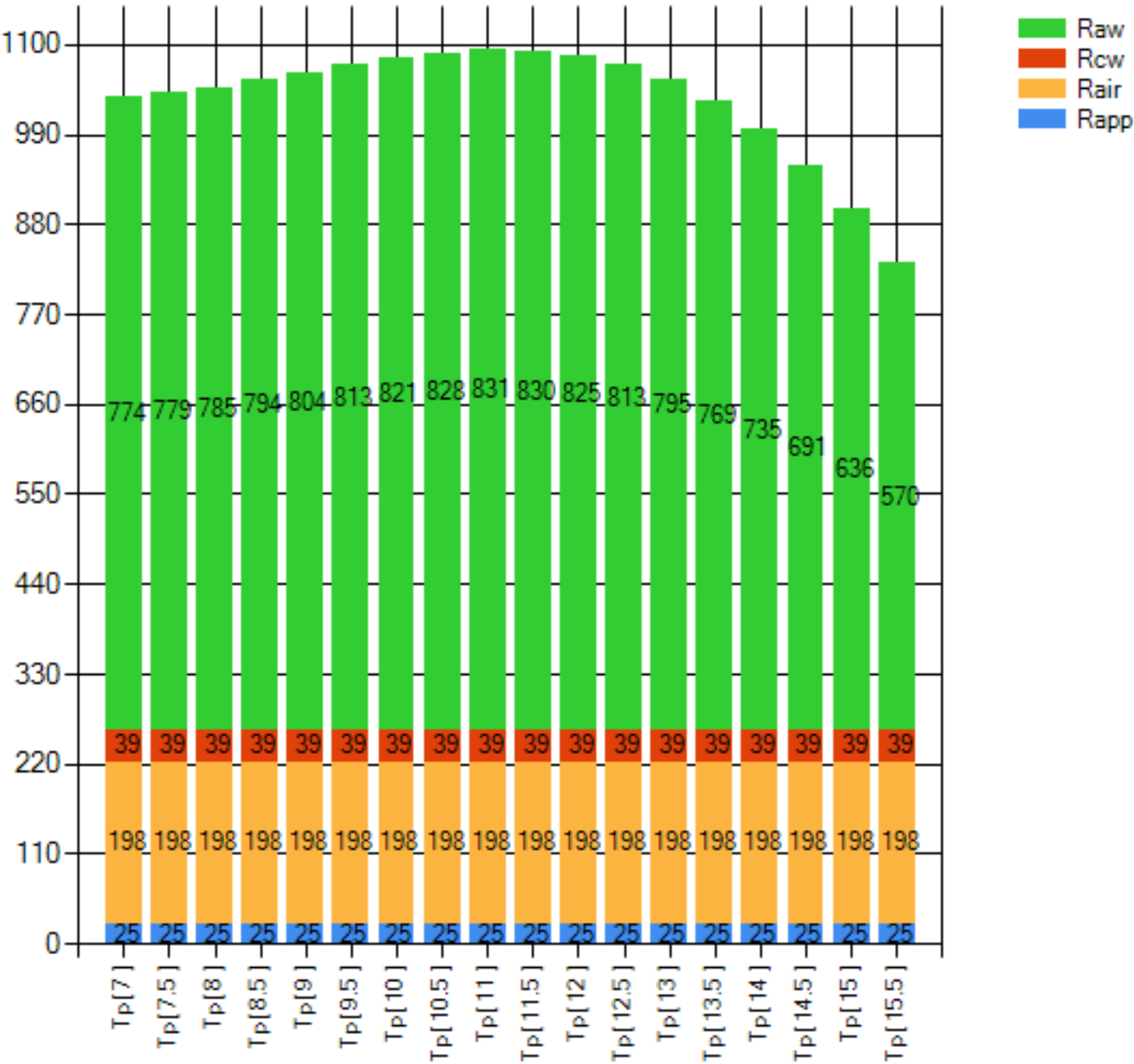
Parameters	Value	Unit
Gravity (g)	9.807	m/s ²
Froude Number (Fr)	0.022534	

Below is the table for Peak wave period (Tps), Added Resistance (Raw), Total Resistance (Rtotal) and Thrust.

TP[s]	Raw[s]	Rtotal[s]	Thrust[s]
7	774.46 kN	1037 kN	1373 kN
7.5	778.6 kN	1041 kN	1379 kN
8	785.48 kN	1048 kN	1388 kN
8.5	794.14 kN	1057 kN	1399 kN
9	803.63 kN	1066 kN	1412 kN
9.5	813 kN	1075 kN	1424 kN
10	821.3 kN	1084 kN	1435 kN
10.5	827.58 kN	1090 kN	1444 kN
11	830.87 kN	1093 kN	1448 kN
11.5	830.25 kN	1093 kN	1447 kN
12	824.74 kN	1087 kN	1440 kN
12.5	813.41 kN	1076 kN	1425 kN
13	795.29 kN	1058 kN	1401 kN
13.5	769.45 kN	1032 kN	1367 kN
14	734.92 kN	997 kN	1321 kN
14.5	690.75 kN	953 kN	1262 kN
15	636 kN	898 kN	1190 kN
15.5	569.71 kN	832 kN	1102 kN

The value of thrust deduction factor tdf obtained from empirical formula is 0.245. By applying the thrust deduction factor, the propeller thrust T can be determined from the total resistance at each peak period.

The stacked graph show below depicts the Total Resistance at each peak wave period.



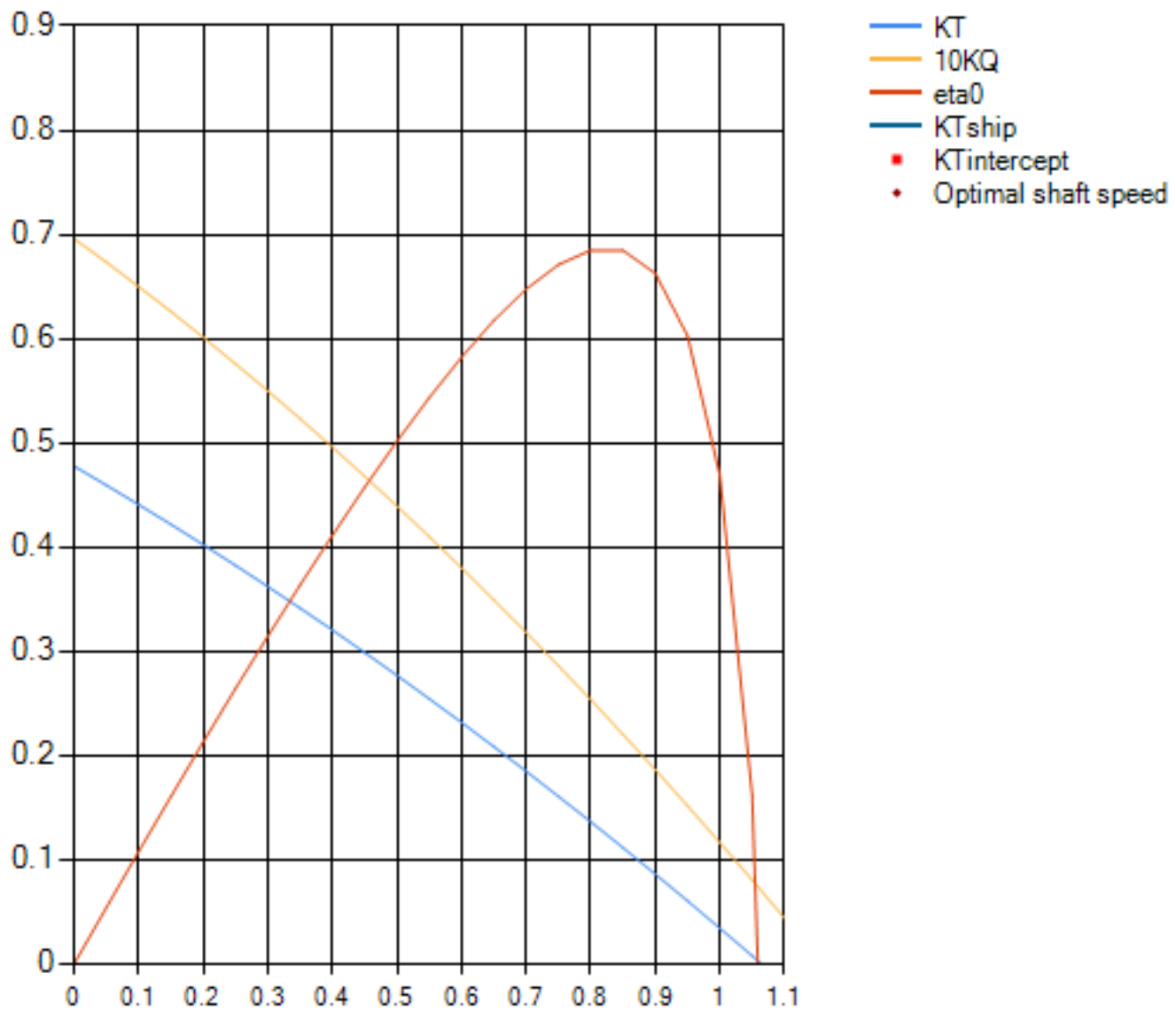
Parameters	Value	Unit
Propeller Diameter (Dp)	7.2	m
Engine Layout	Midship	
Relative rotative efficiency (RReff)	1	
RPM	81.7	rev per minute
Recommended values for wake fraction (w)	0.35	
Thrust deduction factor (tdf)	0.245	
Speed (Ua)	1.34	m/s

From the known required speed of advance and wake, the speed Ua is 1.34 m/s.

The relationship between Advance Coefficient(J), Thrust Coefficient(KT), Torque Coefficient(KQ), Propeller Efficiency(eta0), Optimum propeller speed(nopt) can be illustrated using the table and graph shown below.

J	KT	10KQ	eta0	KTship
0	0.478	0.696	0	NaN
0.05	0.459	0.673	0.054	NaN
0.1	0.441	0.65	0.108	NaN
0.15	0.422	0.626	0.161	NaN
0.2	0.402	0.601	0.213	NaN
0.25	0.383	0.576	0.264	NaN
0.3	0.362	0.55	0.315	NaN
0.35	0.342	0.523	0.364	NaN
0.4	0.32	0.496	0.411	NaN
0.45	0.299	0.468	0.457	NaN
0.5	0.277	0.439	0.502	NaN
0.55	0.255	0.41	0.543	NaN
0.6	0.232	0.38	0.582	NaN
0.65	0.209	0.349	0.617	NaN
0.7	0.185	0.318	0.648	NaN
0.75	0.161	0.286	0.671	NaN
0.8	0.136	0.254	0.685	NaN
0.85	0.112	0.22	0.685	NaN
0.9	0.086	0.186	0.663	NaN
0.95	0.06	0.152	0.603	NaN
1	0.034	0.117	0.469	NaN
1.05	0.008	0.081	0.162	NaN

The interception coordinates for KTship with KT curve are x: NaN and y: NaN.



Thus, the coordinates for Optimal shaft speed are x: NaN and y: NaN.

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